

Full Derivation of Robbins' Formula

- ① Start with a collection of n data points. Each person (i in n) has an unobserved true rate value θ_i and an observed count x_i .

For this, we assume $x_i | \theta_i \sim \text{Poisson}(\theta_i)$

x_i is generated by Poisson distribution centered at θ_i

- ② The prior comes because we have n people each with a different θ_i true rate that varies person to person & in range. So the variation in the θ_i 's means we can say they come from some distribution $g(\theta)$.

$$\theta_i \sim g(\theta)$$

g describes the population of true rates but it's unknown - the EB problem.

- ③ We have $x_i | \theta_i \sim \text{Poisson}(\theta_i)$ and $\theta_i \sim g(\theta)$.

But as an outside observer, you only see x_i . What's the distribution of x_i ?

→ average over all possible values of θ , weighted by how likely each θ is under g :

$$f(x) = \int_0^{\infty} p(x|\theta) g(\theta) d\theta \quad \left. \vphantom{\int_0^{\infty}} \right\} \text{marginal distribution of } x.$$

So $f(x)$ is the probability of observing one count x , taking into account that different people have different true rates θ .

- ④ We observe x_i , but we really care about θ_i , so the question to be answered: given we observe x_i , what's the best estimate of θ_i ?

Standard Bayesian answer → use the posterior mean

$$\hat{\theta}^B(x) = E[\theta | x]$$

* Bayesian estimator under squared error loss, so minimize expected squared error & this is the best you can do.

④ cont.

With substitution/expansion using Bayes' Theorem.

$$E[\theta|x] = \frac{\int_0^{\infty} \theta \cdot p(x|\theta)g(\theta)d\theta}{\int_0^{\infty} p(x|\theta)g(\theta)d\theta} = \frac{N(x)}{f(x)}$$

* So the denominator is $f(x)$ which can be calculated from observed data, but the numerator needs to know $g(\theta)$ to compute θ , which means we're stuck because $g(\theta)$ is unknown. $\rightarrow$ so let's rewrite it.

⑤ Let's focus on the numerator & sub Poisson in for $p(x|\theta)$

$$N(x) = \int_0^{\infty} \theta \cdot \frac{e^{-\theta} \theta^x}{x!} \cdot g(\theta) d\theta$$

Using exponent rules $\rightarrow \theta \cdot \theta^x = \theta^{x+1}$

$$N(x) = \int_0^{\infty} \frac{e^{-\theta} \theta^{x+1}}{x!} g(\theta) d\theta$$

Now if $f(x+1)$ is:

$$f(x+1) = \int_0^{\infty} \frac{e^{-\theta} \theta^{x+1}}{(x+1)!} g(\theta) d\theta$$

then $N(x)$ & $f(x+1)$ are almost identical except the factorial, and $(x+1)! = (x+1) \cdot x!$, so

$$f(x+1) = \frac{N(x)}{(x+1)} \quad \text{rearrange} \quad N(x) = (x+1)f(x+1)$$

which now doesn't need $g(\theta)$ to get an answer.

⑥ Last substitute back to posterior mean:

$$E[\theta|x] = \frac{(x+1) \cdot f(x+1)}{f(x)} \quad \text{which is}$$

Robbins' Formula :

$$E[\theta|x] = (x+1) \cdot \frac{f(x+1)}{f(x)}$$

& unknown prior, $g(\theta)$, is gone!